521326S

RADIO ENGINEERING I

Design Exercise Manual

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(based on previous material
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1. Introduction

In the chapter the definition of the design exercise as well as practical advices for performing the exercise are explained.

1.1. General information

The task is to design and simulate a low-noise preamplifier for a certain bandwidth with certain parameters by using the simulation software uSimmics QUCS-S. First it is beneficial for the student to get familiar with the theory of a small signal amplifier presented in the lectures. Also, the design examples presented in the exercises are useful.

The manuals of the software can be found in electrical format by clicking ***help*** – button. Try using it whenever you face problems. It is recommended that the user without guidance begins the usage of QUCS by reading the QUCS tutorial manual available in Moodle.

1.2. Software to be used

In this design exercise, the free-to-download open source software QUCS (uSimmics) or QUCS-S can be used. The instructions in this document are applicable to both software as the main functionalities are largely identical.

The software can be downloaded to one's own computer from the following links. Note that accessing or using the software does not require the university network or premises.

QUCS/uSimmics: https://qucsstudio.de/download/

QUCS-S: https://ra3xdh.github.io/

2. Design Exercise

2.1. Specifications of the amplifier

Use the BJT transistor model BFP450 during the work. The S-parameter library and transistor datasheet are available on the REI Moodle page. Import the relevant transistor S-parameters into the software with the "S-parameter file" block, where you can specify the location of the Touchstone file.

In the design the goal is to match the transistor into minimum noise figure at frequency band 1.8 – 1.9 GHz in such a way, that the available gain is as high as possible. Using transistor BFP450 at the operating point VCE = 3 V, ICE = 40 mA the following performance values at the whole frequency band 1.8 – 1.9 GHz should be reached:

Noise figure NF: not greater than 1.9 dB
Gain $|S(2,1)|$: not smaller than 13 dB
The transistor must be unconditionally stable within the defined bandwidth.

For the microstrip implementations, consider the following properties for the substrate material: $\varepsilon_r = 4.3$, $h_{sub} = 0.5334$ mm, $t_{metal} = 0.043$ mm, tanδ 0.015

2.2. Using the program

When the schematic is ready and the parameters are set correctly the schematic can be saved and simulated by → ***Simulate***. After the simulation a dataset is created which can be visualized in a new window that usually opens automatically. When the window has opened the correct dataset should be selected from the upper bar. Different kind of graphics can be added in the data display by dropping them by mouse (*Rectangular Plot, Polar, Smith, List* etc.) and the functions displayed in these are selected from the window that is automatically opened.

It is important to check that the data is displayed from the correct dataset. Also graphs from several different datasets (i.e. simulations) can be added to a data display and they can be moved freely by mouse. Also simple graphics and text fields and markers can be added into the data windows to produce clear documents. Clear documents and results should be the aim also in this work.

2.3. Design procedure

During the design, consider the following steps and analyses to characterize the response of the amplifier. It is useful to add markers and texts in the data windows to make the results and further analysis clearer. Remember to save all relevant data windows because they must all be included into the design exercise report. You may for instance proceed in the following order when performing the design exercise:

Design the bias networks for the input and output sides of the transistor. The biasing devices should behave like an open circuit when added to the transistor, and should help maintain stability and satisfies the noise and gain specifications with the transistor. Here, let us use a fan-shaped open-stub circuit that consists of a radial stub, a microstrip line with a length of l/4, and an additional line to connect the DC power source. The fan-shaped microstrip bias line acts as a bypass capacitor in the LNA. It prevents AC leakage caused by insufficient input impedance looking from the RF section to the DC section. On the other way, it is to prevent the DC bias circuit from affecting the impedance characteristics of the AC circuit. Add DC block capacitors to the input and output of the bias networks

Simulate the S-parameters of the transistor with and without the bias networks and compare the results. Are there any major differences or adverse effects seen in the results?. If yes, the dimensions of the network may need adjustments.

Simulate and present in Smith chart (in QUCS data window) the constant noise circles, reflection coefficients S11 and S22 and the reflection coefficient Sopt that is calculated to give the minimum noise figure of the S-parameter transistor model. Present also the gain S21 [dB] and minimum noise figure NFmin [dB] and effective noise resistance Rn.

Examine the stability of the S-parameter transistor model and stabilize it if needed so that it is stable over the whole frequency band of interest ($\mu>1$).

Find out how much the possible stabilizing affects the minimum noise figure and gain. In the data window present also stability factors μ and K and the stability circles of the input and output. The functions mu(), mu2(), rollet(), stabS(), stabL() defined using the equation editor are helpful for completing this task,

Match the transistor using lumped components to reach minimum noise figure in such a way that the specifications for the task are fulfilled

Simulate the whole amplifier you have designed and present the obtained values of noise figure, gain and reflection coefficients S11 and S22. In simulation, use the largest possible bandwidth, provided by the S-parameter data of the amplifier.

Voluntary bonus task 1:
Do the matching by using microstrip lines and compare the result with the result obtained by lumped components. This task is not compulsory and it can only raise the grade of the laboratory work report. In practise use some of the default microstrip line components in QUCS. You can determine the electrical lengths of the stripes either by calculating them theoretically by hand or by using QUXS. Finally, build a similar circuit as with the lumped components in order to be able to simulate the performance of the strip solution. Realize only the matching components with the strips.

Voluntary bonus task 2:
After finalizing the structure, check the stability containing the amplifier, stability circuit, and the input and output matching circuits.

Voluntary bonus task 3:
Discuss the possible changes in your design in the previous steps if the operating frequency of the amplifier should be much higher, e.g., on the order of 10 GHz.

3. PRINTING THE RESULTS

You can print the schematics and data displays into files in different formats. Then attach the figures into the Design Exercise Report.

In order to print to a file select ***File->Print*** and select "***Create File***". Please check that the figure in the produced file looks like it should be.

4. DESIGN EXERCISE REPORT

The design exercise report's structure and appearance is supposed to apply the department's instructions for the Master's Thesis. The instructions can be found in Finnish from:

https://unioulu.sharepoint.com/:b:/s/Liitetiedostot-Attachments/EbMNDuR_FxJDiAm4TSRExJsBt2opXr5miZo42JPXzkxoBw?e=dDgXrX

and in English from:

https://unioulu.sharepoint.com/:b:/s/Liitetiedostot-Attachments/ESOZUibmhPtBjRNDjVsEcpwBBWmGOwwLUu86Rpn67AvmJA?e=SckfNp

There is also a template document in Word format that has the same page settings as described in the department's instruction document above. It can be found in Finnish from:

https://unioulu.sharepoint.com/:w:/s/Liitetiedostot-Attachments/EcouTpL_FiZElHRJgZnVV7wB9D48bFODqNptTqPCGoiJfQ?e=qrel8C

and in English from:

https://unioulu.sharepoint.com/:w:/s/Liitetiedostot-Attachments/EYiyBQM9jlRFiE3R0tFMUYsB1WGX248DfLTBErsi7dplMw?e=ZLnfoI

and in Latex-format

https://www.overleaf.com/read/spsnwyhdccff

In the design exercise report at least the following things should be included:

- The essential parts of the theory needed during the course of the design exercise.
- All schematics and data windows used/produced in the simulations. The results must be clearly displayed, use the markers.
- Matching of the input and output also performed manually in Smith chart/paper with calculations. Justifications for reflection points in matching the input and output must be clearly presented.

- Comments and considerations on the simulation results and performing the work. Selected choices during the work must be justified.
- Feedback and notes on the design exercise on general.

The design exercise will be done in groups of up to three students. If doing the work in groups, please indicate the contribution of each student to the overall work. Also mark down an estimate of how many hours it took to complete the exercise (including , e.g., reading relevant parts of the course literature).

The deadline for returning the report is 10 January 2026. Each group returns one report and the reports must be individual between the groups. The reports are returned electronically by e-mail in pdf-format.

The reports will be evaluated all at the same time, within three weeks period of time after the deadline.

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